

APURBO BANIK TURJO

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SUMMARY

Software Engineer with **1+ year of industry experience** shipping production features on large-scale **Java microservices**, with a growing research focus on **on-device machine learning and efficient AI**. Hands-on in **agentic AI development**, LLM-integrated workflows, and cross-platform full-stack development. Drove **WCAG 2.2 AA** accessibility compliance in production and integrated compression in content response for a service. Undergraduate thesis on **dengue breeding site detection** achieving **83.6% balanced accuracy**, under review. Currently working on how different **model compression** techniques impact the fairness and trustworthiness, with the goal of enabling **efficient, fair, and reliable models** for resource-constrained environments.

EDUCATION

Bangladesh University of Engineering and Technology

B.Sc. in Computer Science and Technology; **CGPA: 3.8732/4.00**

Relevant Coursework: DSA (I, II), OOP, System Design, Database, OS, ML, Compiler, etc.

Awards: Dean's List (2 semesters)

Dhaka, Bangladesh

Feb 2020 – March 2025

Shahed Syed Nazrul Islam College

Higher Secondary Certificate; **GPA: 5.00/5.00**

Mymensingh, Bangladesh

2019

SKILLS

Languages: C/C++, Java (8, 21), Python, JavaScript, TypeScript, SQL, Bash

Frameworks & Libraries: Spring Boot, Hibernate, React, React Native, Capacitor, Express, Django

Tools & Platforms: Git, Docker, Nginx, Jenkins, Postman, Figma, n8n, MCP Server, Hazelcast, Kafka

Databases: PostgreSQL, MySQL, OracleDB, Supabase

Cloud & Hosting: Azure, AWS, Firebase, Render, Vercel

AI & ML: Agentic Development, LLM Integration, Model Compression, Fairness Evaluation, Federated Learning

WORK EXPERIENCE

Software Engineer, Therap BD Ltd.

Core Product Team (Java Microservices)

Dhaka, Bangladesh

Apr 2025 – Present

- Dockerized a core module for **multi-instance availability** in the dev environment; integrated **compression** into the request-response pipeline of a core service, reducing payload sizes and improving response throughput.
- Drove **WCAG 2.2 AA** accessibility compliance across a core module (ARIA roles, keyboard navigation, screen reader semantics); conducted R&D on **RabbitMQ** and **NATS-JetStream** for event-driven architecture evaluation.
- Built an **AI-powered alert summarization pipeline** with **n8n** that processes production system alerts into actionable digests; developed a **Google Chat bot** (adopted by the team).
- Contributed to a **guided test automation framework** using **Gemini CLI and MCP servers**, streamlining UI regression testing for new features and bug fixes; practiced **agentic development workflows** extensively for accelerated feature delivery.

RESEARCH EXPERIENCE

Undergraduate Thesis Research

Bangladesh University of Engineering and Technology

Dhaka, Bangladesh

2023 – 2025

- **Title:** “Detection of Dengue Breeding Sites in Large-scale Landscapes Plagued by Unplanned Urban Development from Aerial Imagery and Remote Sensing with Deep Learning”
- Fine-tuned vision models on a custom dataset tailored to Bangladesh’s unique urban structures, achieving **83.6% balanced accuracy** in automated detection of high-risk buildings from UAV orthophotos.
- Developed a novel **score-based OBB merging algorithm** for precise building classification and geo-location identification of risky zones across large-scale orthophotos.

- Integrated multiple dengue breeding factors with a **distance-based clustering strategy** to mirror real-world epidemiological patterns.
- Enabled **35% reduction in health monitoring costs** through targeted resource allocation; applicable to similar unplanned urban topographies across developing regions.
- **Status:** Manuscript under review | Backed by EDGE Project BD | **Tools:** Python, MATLAB, YOLO, SegGPT, QGIS

Ongoing Research Project

Dhaka, Bangladesh

Collaboration with Professor from BUET

2025 – Present

- Investigating model compression techniques (**pruning, quantization, knowledge distillation**) for edge deployment on resource-constrained devices.
- Analyzing trade-offs between **model efficiency, performance, and trustworthiness**; evaluating fairness implications of compressed models across diverse populations.

PROJECTS

LearnHub; Auth Server [Spring Boot 3, Java 21, Kafka, Hazelcast, Docker] | [GitHub](#)

- Designed a **JWT microservice** (RSA-256 asymmetric signing, refresh token rotation) with a dedicated **key provider microservice** enabling decentralized, stateless token verification across services; built a reusable **authcommon** library with a pluggable servlet filter, centralized expiry/blacklist validation, and **TTL-based public key caching**; eliminating redundant key fetches across consumers.
- Implemented **role-based access control** (Admin / Instructor / Student) with admin-controlled account toggling and **real-time force-logout via Hazelcast distributed cache** eviction; used **Apache Kafka** for async event publishing (registration, deletion) to downstream services; packaged in a **Gradle multi-project build** with Docker and global exception handling with i18n-ready error messages.

CostPilot; Personal Finance App [React 19, TypeScript, Vite, Capacitor 8, jsPDF] | [GitHub](#) | [Live](#)

- Built a **local-first cross-platform finance app** (Web + Android) offering typically premium features at zero cost: reusable budget plans with threshold alerts, custom category taxonomy, branded PDF/CSV data exports, calendar heatmap, and dual-mode spending analytics; deployed from a single React/TypeScript codebase to both Vercel and a signed Android APK via Capacitor.
- Designed an offline-capable **rolling backup system** with scheduled auto-backups, missed-window catch-up on app resume, same-day overwrite conflict detection, content-hash deduplication to skip redundant writes, and a merge-on-restore flow that intelligently reconciles backup data with live device state without any external sync dependency.

Screen Arxiv; Media Tracking App [React 19, React Native 0.84, TypeScript, TMDB API] | [GitHub](#) | [Live](#)

- Built a **dual-platform media consumption awareness app** (React/Vite web + React Native Android) designed to surface how much screen time is spent and on what; features status-based tracking (Watched, Watching, Waitlist), granular per-episode TV progress with bulk actions, upcoming episode/movie release schedules, and a **library analytics engine** with decade distribution and genre breakdowns to quantify consumption patterns.
- Integrated TMDB API with real-time debounced search and auto-fetched metadata (cast carousels, trailers, release dates); designed a layered modal navigation system for hierarchical content browsing; implemented gesture-based swipeable tabs, JSON backup/restore for full library portability, and adaptive theming across both platforms from shared business logic.

Predictive Diagnosis System [Python, BioBERT, NLP, Medical AI] | [GitHub](#)

- Fine-tuned **BioBERT** on preprocessed **DDXPlus** dataset to build personalized diagnostic profiles from medical history, symptoms, and demographics.
- Generated **probabilistic disease predictions** enabling early intervention and clinical decision support.

EduByte [React, Bootstrap, Express, PostgreSQL] | [GitHub](#) | [Demo](#)

- E-learning platform with structured CS course content, an **interactive AI chatbot**, automated assessments, progress tracking, and a collaborative discussion forum with posts, reactions, and comments.

AWARDS & ACHIEVEMENTS

Dean's List (2 semesters): Awarded for consecutive levels at BUET for academic excellence.

National Quiz Competition: Advanced to nationals in high school quiz competition organized by Shishu Academy Bangladesh.

ADDITIONAL INFORMATION

Languages: Bengali (Native), English (Fluent)

Interests: Reading, Running, Swimming, Fitness, Table Tennis